

Joe Hollander

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EDUCATION

Columbia College, Columbia University

B.A. Mathematics

New York, NY

Expected 2030

Relevant Ongoing Coursework: Honors Mathematics

South Burlington High School

South Burlington, VT

GPA: 3.99/4.0 SAT: 1580 PSAT: 1520/1520

2022–2026

Relevant Coursework: AP Calculus BC (5); AP Statistics (5); Calculus III and Applied Linear Algebra (UVM Dual Enrollment)

AWARDS & COMPETITIONS

- **Vermont State Math League Champion**, four consecutive years — undefeated as team captain in 2026
- **2nd place individual**, Vermont State Math League; **1st place**, Advanced Math division
- **1st place statewide**, University of Vermont High School Prize Examination
- **ARML invitee** — 1/16 selected for Vermont's team

RESEARCH

Gold Futures and Leveraged-ETF Replication — independent research paper: [rough draft](#) •

- Tested whether futures portfolios track spot gold and leveraged ETFs better than the ETFs track their own benchmarks
- Built continuous futures series from raw CME data, implemented two roll methodologies (time and volume based), back-adjusted for splicing, and validated against an independent reference
- Solved optimal replicating portfolios as a constrained quadratic program (`cvxpy`); several beat **GLD's own out-of-sample tracking error**
- Found **statistically significant** leverage deviations in **2 of 4** ETFs via t-tests on regression slope and daily tracking bias
- Created a dynamic (daily-rebalanced) replication strategy using real EFFR data; analyzed regime-dependence (especially volatility) of tracking error and leverage deviation
- Produced draft of findings using $\text{\LaTeX}$ and `matplotlib`, including a variety of visualizations and tables

PROJECTS

Options Volatility and Flow Analysis

- Replicated the CBOE VIX methodology from white paper end-to-end on AAPL options, including put-call parity forward extraction and cubic-spline interpolation of Treasury rates
- Priced options via various methods (Black-Scholes, Cox-Ross-Rubinstein, Leisen-Reimer); solved implied volatility with Newton-Raphson and analytic vega;
- Engineered signals from open-interest concentration and bid/ask-size imbalance across moneyness and time-to-expiration to visualize option flow pressures relative to price action.

Machine Learning Competitions

- **Top 25%** of 3000+ entrants, Kaggle Podcast Listening Time
- Applied cyclical encoding, validated model-based imputation, and leakage-safe target encoding via nested cross-validation
- Implemented model-based (XGBoost) imputation for missing fields and brute-force feature combination
- Designed a leakage-safe nested K-fold target-encoding scheme (mean/quantile/histogram target statistics) for categorical predictors

Crypto Tick Data Analysis

- Containerized a FastAPI service streaming multi-exchange order-book data into ArcticDB
- Computed a cost-based VWAP walking book levels with partial-fill interpolation to estimate execution price at a target notional

SKILLS

- **Languages & Tools:** Python 3, $\text{\LaTeX}$, Git, Docker, asyncio, ccxt, Databento
- **Scientific Computing & ML:** NumPy, pandas, SciPy, statsmodels, cvxpy, Matplotlib, scikit-learn, XGBoost, LightGBM

ACTIVITIES

- Math League (Captain 2026); Math Tutoring; Ping Pong Club President (2024-2026); Varsity Ultimate Frisbee; Varsity and Club Soccer

Soccer Referee — Vermont Soccer Association

- **600+ hours** officiating, center refereeing through the **U19** level; selected as a **State Cup referee**
- Demands split-second judgment under pressure, sustained focus, and conviction in a call